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RIVET-A3 Multi-Key, Multi-Context AES Cryptographic Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:16 UTC+8 Beijing Time)

Abstract

RIVET-A3 is a standalone SystemVerilog cryptographic subsystem supporting AES-128/256 ECB, CTR and GCM through legacy streaming, direct descriptors and descriptor DMA. Four hardware keyslots and four saved contexts enable interleaved GCM sessions without software handling intermediate cryptographic state. Authentication-gated payload retention prevents failed GCM decryption from releasing plaintext. The behavioral campaign passed all nine combinations of capacities 1, 3 and 16 blocks with seeds 17, 89 and 203. Additional checks exercised the 256-block maximum, rejected illegal capacities, behaviorally detected a disposable cryptographic mutation and reproduced a focused campaign from copied deliverable sources. Default-capacity generic synthesis reported 2,200,117 generic primitive cells and zero inferred latches. These results establish bounded RTL functionality and generic synthesizability, not physical implementation or security certification.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

A shared iterative AES engine and a 128-cycle polynomial GHASH multiplier serve a byte-serial session controller. Sharing avoids four replicated cryptographic engines while preserving arbitrary-byte GCM fragment boundaries. Four saved records preserve continuation state; separate payload banks retain output and destination ownership across descriptors. The scheduler owns DMA queues, descriptor fetching, memory transactions and completion delivery. Fully resettable storage favors explicit zeroization over technology-specific RAM inference. Blocks are arranged row-major in three columns; relationships represent concurrent control, data and ownership paths, not a serial pipeline. Architecture and implementation are documented by E001, E002 and E014.

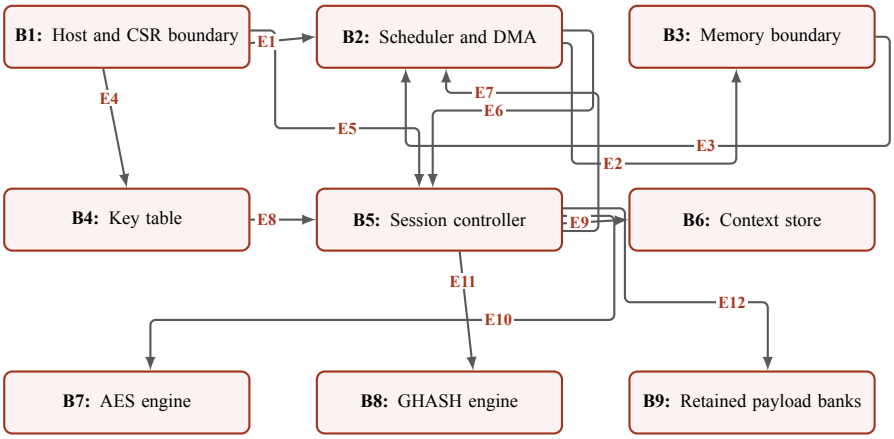


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Host and CSR boundary	Legacy commands, streaming data, direct descriptors, DMA submissions, key operations and status registers.
B2: Scheduler and DMA	u_scheduler_dma implements independent depth-two high/normal FIFOs, descriptor dispatch, aligned memory requests, status writes and retained completions.
B3: Memory boundary	External 128-bit request/response interface with 32-bit byte addresses and exactly one outstanding accepted read or write.
B4: Key table	u_key_table stores four normalized keys with independent length, validity and reset-lifetime lock; bound or locked replacement is rejected.
B5: Session controller	Validates descriptors, restores contexts, consumes bytes, sequences cryptography, finalizes results and scrubs consumed state.
B6: Context store	u_context_store contains four 1280-bit records with combinational reset-gated reads and edge-committed clear/write operations.
B7: AES engine	u_aes_core contains adapted key-schedule, datapath, S-box, ShiftRows and MixColumns modules; supplies ECB transforms, keystream, H and tag masks.
B8: GHASH engine	u_ghash performs polynomial multiplication for ciphertext authentication and aggregate-length processing.
B9: Retained payload banks	Four context-owned byte banks and per-byte destination maps retain enhanced output until finalization and retirement.

Directed relationship	Meaning
E1: B1 to B2	Descriptors, submissions and completion handshake
E2: B2 to B3	Memory requests
E3: B3 to B2	Memory responses and faults
E4: B1 to B4	Atomic key loads and result consumption
E5: B1 to B5	Legacy commands, streams and CSR control
E6: B2 to B5	Descriptor dispatch and status-fault notification
E7: B5 to B2	Byte I/O, results and scrub acknowledgement
E8: B4 to B5	Selected key, length and validity
E9: B5 to B6	Context selection, commit and clear
E10: B5 to B7	Key/block sequencing and AES result consumption
E11: B5 to B8	Hash operands and product consumption
E12: B5 to B9	Retain, authenticated drain and scrub

Signal / group	Direction	Width	Contract

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Single clock and asynchronous active-low reset. Public outputs are zero during reset; availability begins after the first release rising edge.
Key channel	Input/output	256-bit key; 2-bit slot; 4-bit result	key_valid/key_ready accepts key_in, key_len_sel, key_slot_id and key_lock. Length select 0 uses key_in[255:128] for AES-128 and clears the unused half; 1 selects AES-256. key_result_valid/code remain until key_result_ready. key_loaded reflects slot0 validity.
Legacy raw ECB	Input/output	128-bit block; 8-bit transaction ID	in_valid/in_ready transfers in_block, in_enc_dec and in_tag_id using slot0. Shared output echoes ID and direction. Unloaded attempts assert err_unloaded_key without accepting work.
Legacy GCM command and streaming input	Input/output	96-bit IV; two 32-bit lengths; 128-bit tag/data; 5-bit byte count	cmd_valid/cmd_ready accepts cmd_decrypt, iv_in, aad_bytes, payload_bytes and expected_tag. data_valid/data_ready transfers data_is_aad, data_in and valid_bytes, AAD before payload. Each beat supplies exactly min(16, remaining section bytes) MSB-valid bytes.
Shared payload and legacy tag outputs	Output with input readiness	128-bit payload/tag; 8-bit ID; 5-bit byte count	out_valid/out_ready carries out_block, out_tag_id, out_enc_dec, out_valid_bytes and out_is_gcm. Unused bytes are zero; enhanced output ID is zero. tag_valid/tag_ready carries tag_out; tag_match, auth_fail and protocol_error describe the legacy/shared result. Legacy GCM retires its tag before payload.
Direct task descriptor	Input with output readiness	512 bits	task_valid/task_ready accepts task_descriptor. Direct jobs ignore descriptor addresses and use streaming input; enhanced and legacy input ownership cannot mix.
DMA submission and completion	Input/output	32-bit descriptor address; 8-bit cookie; 2-bit context; 4-bit code; 128-bit tag	dma_submit_valid/ready accepts dma_desc_addr, dma_priority and dma_cookie; priority 1 selects high. completion_valid/ready retains completion_cookie, completion_ctx_id, completion_code, completion_final and completion_tag. DMA uses the submission cookie; direct work uses descriptor byte56.
Memory request/response	Output requests; input responses	32-bit address; 128-bit data; 16-bit write strobes	mem_req_valid/ready transfers mem_req_write, mem_req_addr, mem_req_wdata and mem_req_wstrb. mem_rsp_valid/ready transfers mem_rsp_rdata and mem_rsp_error. Requests are 16-byte aligned; lane i maps address+i to bits[8*i+:8]. Reads have zero data/strobes. Every accepted request, including writes, requires one response before another request.
CSR and activity	Input/output	8-bit address; 32-bit write/read data; scalar control	csr_valid/ready accepts csr_write, csr_addr and csr_wdata; csr_rsp_valid/ready retains csr_rdata and csr_error. Only one response is outstanding. busy includes active, queued, scrub and reserved-output work, but not merely suspended contexts.

Parameter	Default	Meaning / supported values
MAX_PAYLOAD_ BLOCKS	16	Integer 1..256; 0 and 257 receive explicit parameter diagnostics. Each context retains 16 times this value in payload bytes: 256 bytes by default and 4096 maximum. Four contexts retain 1024 default payload bytes plus 4096 destination-mapping bytes. Context records total 640 bytes and keys total 128 bytes. These are logical capacities, not physical area.

III Behavior and corner cases

A Handshake and ordering

Transfers require valid and ready immediately before a rising edge; effects appear afterward. Stalled producers hold all fields stable. Newly produced responses cannot retire on their creation edge. Retirement releases dependent ownership afterward. Simultaneous queue enqueue/dequeue executes the old head and preserves FIFO order.

B Descriptor and status encoding

The 64-byte descriptor places byte i at descriptor[8*i+:8]. LE32 fields at offsets 0/4/8/12/16/20/24 hold control/ AAD address/source/destination/status/AAD count/payload count. Bytes28..39 hold chronological IV, 40..55 expected tag, 56 cookie and 57..63 zero. Control contains mode1:0 (ECB0, CTR1, GCM2), decrypt2, context4:3, slot6:5, first7 and last8; other bits are zero. Status occupies 32 bytes: LE32 code and cumulative payload count, cookie8, context9, final10, zero11..15 and chronological tag16..31. Tags are zero except successful final GCM.

C Cryptographic arithmetic

Crypto byte0 occupies bits127:120. ECB supports encryption/decryption, no AAD and whole-block lengths, including empty work. CTR uses AES encryption in either direction, begins at IV||1 and increments only the low32-bit big-endian counter. GCM uses J0=IV||1, first payload counter IV||2, H=AES_K(0), ciphertext GHASH and aggregate64-bit big-endian bit lengths. All128 tag bits participate in verification. Counter reuse and length overflow are rejected.

D Session continuity

ECB/CTR require first and last together. GCM supports first/middle/last and empty fragments at arbitrary byte boundaries. Contexts preserve binding, mode, direction, origin, IV, GHASH accumulator, H, mask, counter, lengths, pending hash bytes, unused keystream and retained count. Padding occurs at the AAD-to-payload transition and finalization, not each fragment. Nonempty AAD after payload begins is invalid. Invalid continuations preserve saved state; errors after consumption scrub the affected session.

E Scheduling and authenticated retirement

Idle precedence is accepted key operation, high DMA, normal DMA, direct task, legacy GCM, then raw ECB. Active descriptors are nonpreemptive. Nonfinal state commits before completion. All enhanced output is retained until finalization. Direct completion retires before payload emission, retaining ownership through final output. DMA completion follows acknowledged payload writes and status delivery. Authentication failure releases no payload and performs no payload write.

F Memory safety and faults

Descriptor/status addresses require16-byte alignment; accessed ranges cannot wrap32-bit addresses. Exact in-place payload is supported; prohibited overlaps are rejected before payload writes. Zero-length sections generate no section accesses. Byte transactions preserve unaligned destinations using exact strobes. Memory faults return code8 without retry; unknown status addresses are not invented after fetch failure. Status-write faults remain observable through completion. Acknowledged writes cannot be rolled back.

G Keys, CSR errors and zeroization

Locks persist until reset; bound or locked writes fail atomically. Codes0..8 mean OK, invalid descriptor/register, invalid key, locked/bound, invalid context, capacity/length, stream protocol, authentication failure and memory error; 9..15 are reserved. CSR offsets0x00/04/08/0c/10/14 provide ID0x52415633, busy, valid/lock masks, live mask, sticky W1C errors and guarded context abort. Unsupported, misaligned, permission-invalid or reserved-bit accesses fail. Abort rejects active/queued/reserved references, including any queued DMA, and scrubs before responding. Reset cancels work without completion and zeroizes storage. Reset dominates clear, clear dominates context write, and new errors dominate simultaneous W1C.

IV Verification methods and measured results

A Independent cryptography and tool environment

Method. Byte-oriented oracle independent of DUT tables/control, anchored by FIPS197 AES and published NIST GCM answers; retained tool-version and runner-preservation checks.

Result. Four AES and ten GCM known-answer comparisons passed. Recorded tools were Python3.12.3, Icarus Verilog/vvp12.0 and Yosys0.33. Fourteen evidence-runner regression checks passed.

Evidence: E021, E023, E1062

B Full seeded cross-feature behavior

Method. Compiled RTL simulation comparing payload bytes/counts/padding, tags, cookies, contexts, codes, memory traffic and absence of extra output, with strict public-output X/Z rejection.

Result. All9/9 capacity-seed combinations passed. For seeds17/89/203 respectively, comparison counts were4886/4861/4822 at capacity1, 5042/5115/5055 at capacity3 and5978/6036/5948 at capacity16. Each run recorded12 bus faults,131 authentication rejections,42 reset checks and a simultaneous queue event. The complete ledger is evidence/mandatory-rerun-001/behavior/receipt.json. Recovery repeated the same matrix, not additional configurations.

Evidence: E020, E022, E1063, E1069

C Capacity boundaries and rollover semantics

Method. Integrated simulations at capacities1/3/16/256, illegal-parameter runs, public unit interfaces and independently checked boundary arithmetic.

Result. All four legal configurations passed; 0 and257 were explicitly rejected. Targeted capacity256 behavior performed18,250 comparisons over716,139 clock edges. Unit checks passed106 comparisons, including80 arithmetic cases and four context banks. Capacity256 coverage was targeted, not the full seeded matrix; arithmetic checks did not transmit enormous messages.

Evidence: E019, E1064, E1070

D Substantive integrated synthesis and zeroization structure

Method. Yosys hierarchy checks, generic synth -noabc, consistency assertions and complete netlist inspection, including required named instances and asynchronous-reset storage connections.

Result. Capacity16 synthesis reported2,200,117 generic primitive cells, zero inferred latches and no unresolved cells; recovery reported the same primitive total. The verification helper's2,200,127 JSON cell entries additionally include ten hierarchical module instances. Checked resettable storage included40,960 top-level payload/mapping bits,5120 context bits and1024 key bits. These are structural results, not physical area or a noninterference proof.

Evidence: E024, E1065, E1071

E Complete evidence retention after capture failure

Method. Lossless gzip retention with decompressed size/hash verification and self-inclusive bundle accounting; fresh exclusive outputs preserve earlier failures.

Result. An earlier1,698,395,392-byte bundle exceeded the1,073,741,824-byte capture limit. The repaired current synthesis bundle measured67,204,919 bytes and passed capture. Complete hierarchy/netlist JSON compressed to19,699,202/34,484,933 bytes with verified round trips. Output collisions were resolved through fresh allocations, not evidence deletion.

Evidence: E033, E035, E023, E1065

F Checker sensitivity and portable reproduction

Method. Compile and simulate a disposable XOR-one transformed-byte mutation; separately copy only declared deliverables and perform fresh compilation plus focused cross-feature simulation.

Result. The mutation compiled and was behaviorally rejected by a GCM completion-tag mismatch at capacity1, seed17. Clean reproduction passed the same capacity/seed focused campaign with4886 comparisons, without the reference mount, prebuilt binaries or private state. This was not a fresh full matrix or physical-flow reproduction.

Evidence: E1066, E1067, E1072, E1073

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	6 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 10	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

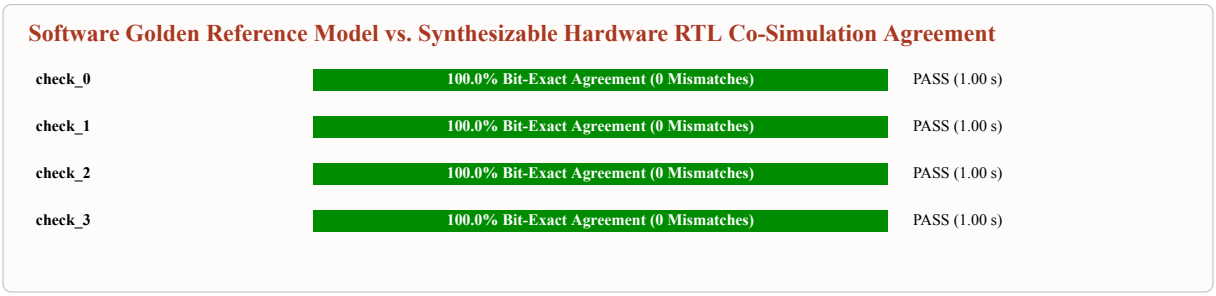


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0

VI Integration and use

Use rtl/*.sv, design/SPEC.md, design/ARCHITECTURE.md and tests/* as deliverable inputs. Compile rivet_a3_pkg.sv before the other eleven RTL files and select rivet_a3_top. Adapted AES primitives are included; the read-only A2 reference is provenance, not a build dependency.

Assert reset low, keep input valids low, release reset and wait for post-release readiness. Load required keyslots and consume key results before dependent work. A suspended context binds its key until final retirement or successful abort.

Keep descriptors and source data stable while owned and prevent writable overlap between outstanding jobs. Supply one response for every accepted memory request, discard pre-reset responses and provide finite back-pressure and response delays.

Use Python3.9 or newer, iverilog, vvp and Yosys on PATH. A focused clean reproduction is invoked from the delivered root by python3 tests/verify.py --suite reproduce --output evidence/recipient-reproduce-001; the destination must not exist. Suite commands are documented in design/SPEC.md and tests/README.md. Allocate fresh destinations rather than reusing recorded output paths.

The synthesis helper handles Yosys0.33 package-import limitations by inserting the verbatim package body into generated inputs; functional simulation uses unchanged package/import RTL. Compressed complete netlists, raw logs and source-bound receipts reside under the recorded suite outputs.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 12 Verilog/SystemVerilog files; 1548 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_a3_aes_core.sv	48 total	45 code
rtl/rivet_a3_context_store.sv	22 total	22 code
rtl/rivet_a3_datapath.sv	86 total	81 code
rtl/rivet_a3_ghash.sv	63 total	61 code
rtl/rivet_a3_key_schedule.sv	90 total	86 code
rtl/rivet_a3_key_table.sv	59 total	59 code
rtl/rivet_a3_mixcolumns.sv	24 total	23 code
rtl/rivet_a3_pkg.sv	105 total	97 code
rtl/rivet_a3_sbox.sv	15 total	14 code
rtl/rivet_a3_scheduler_dma.sv	263 total	263 code
rtl/rivet_a3_shiftrows.sv	17 total	16 code
rtl/rivet_a3_top.sv	783 total	781 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E014	rtl/rivet_a3_top.sv
E019	tests/boundary_units.sv
E020	tests/bridge.sv
E021	tests/oracle.py
E022	tests/scenarios.py
E023	tests/test_runner.py
E024	tests/verify.py
E033	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E035	.vulcan/engineering/run-0001/attempt-0008/stdout.log
E1062	.vulcan/engineering/run-0002/attempt-0003/stdout.log
E1063	.vulcan/engineering/run-0002/attempt-0004/stdout.log
E1064	.vulcan/engineering/run-0002/attempt-0005/stdout.log
E1065	.vulcan/engineering/run-0002/attempt-0006/stdout.log
E1066	.vulcan/engineering/run-0002/attempt-0007/stdout.log
E1067	.vulcan/engineering/run-0002/attempt-0008/stdout.log
E1069	.vulcan/engineering/run-0002/attempt-0010/stdout.log
E1070	.vulcan/engineering/run-0002/attempt-0011/stdout.log
E1071	.vulcan/engineering/run-0002/attempt-0012/stdout.log
E1072	.vulcan/engineering/run-0002/attempt-0013/stdout.log
E1073	.vulcan/engineering/run-0002/attempt-0014/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).